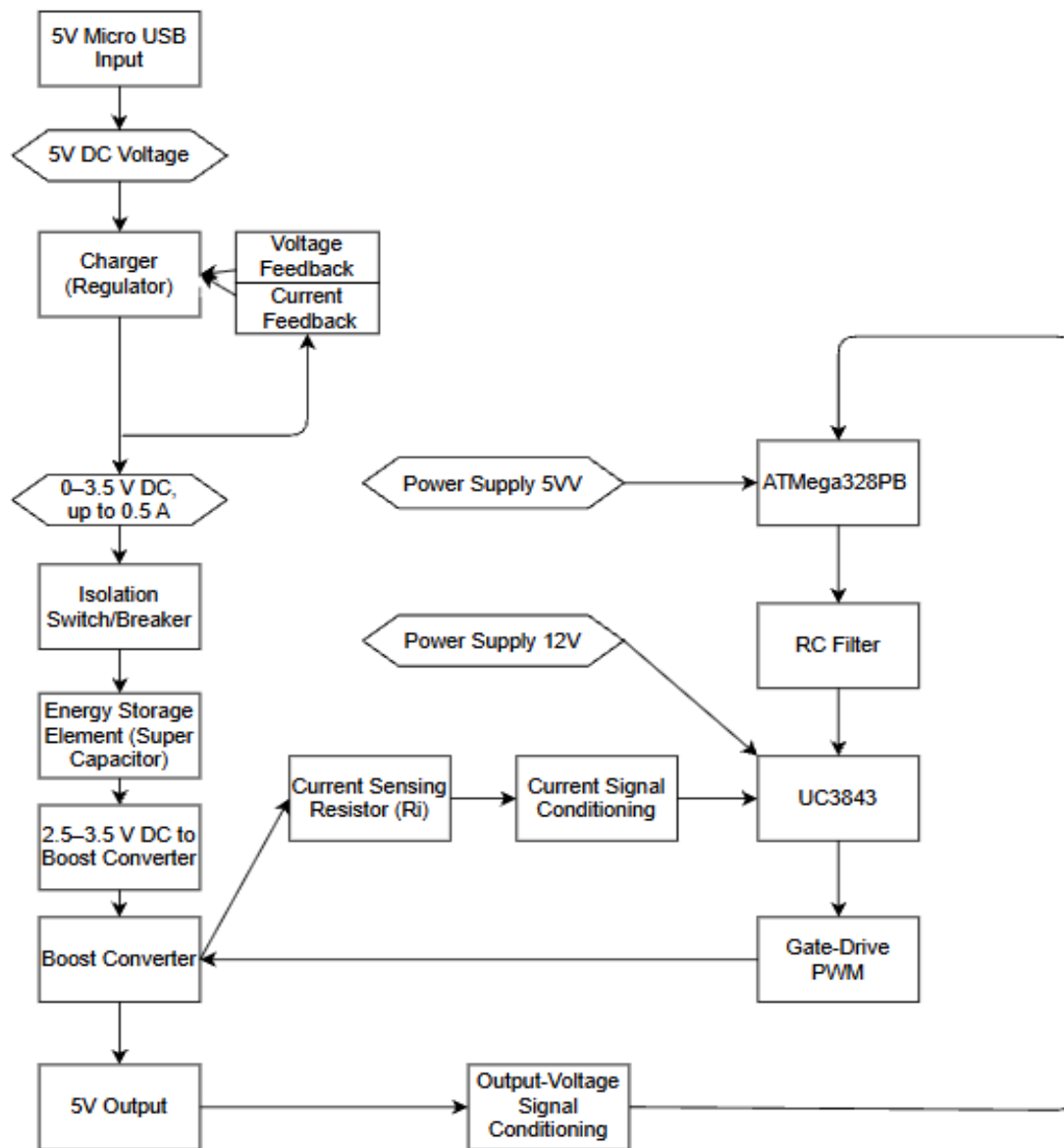
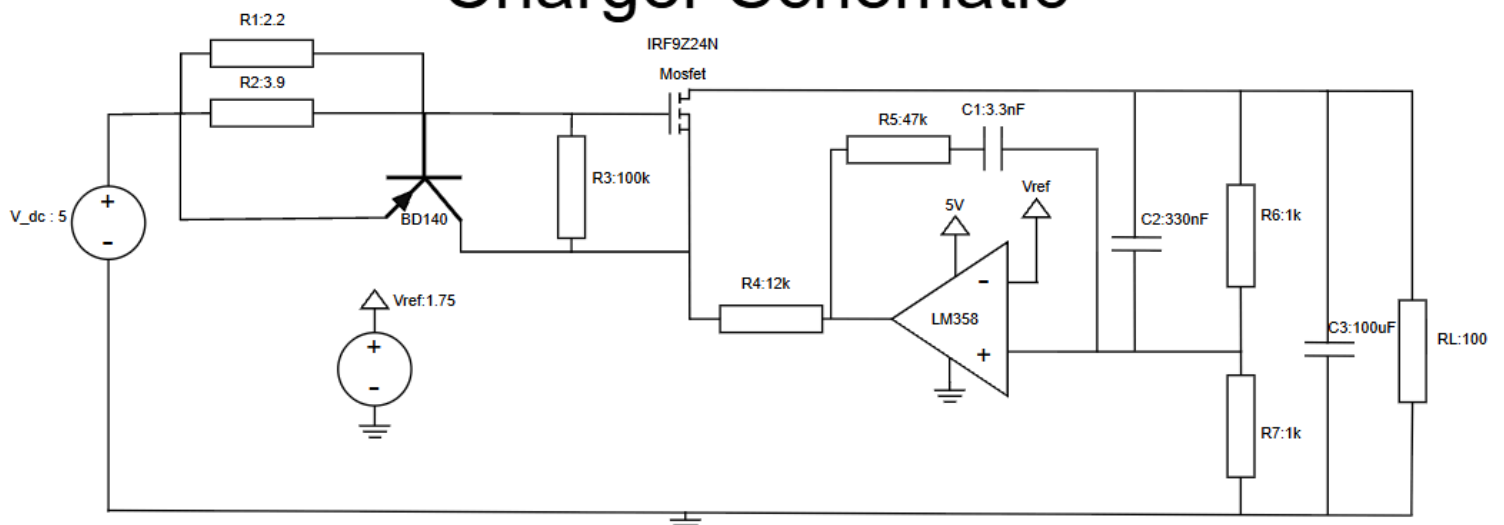


Project Proposal - Group 9



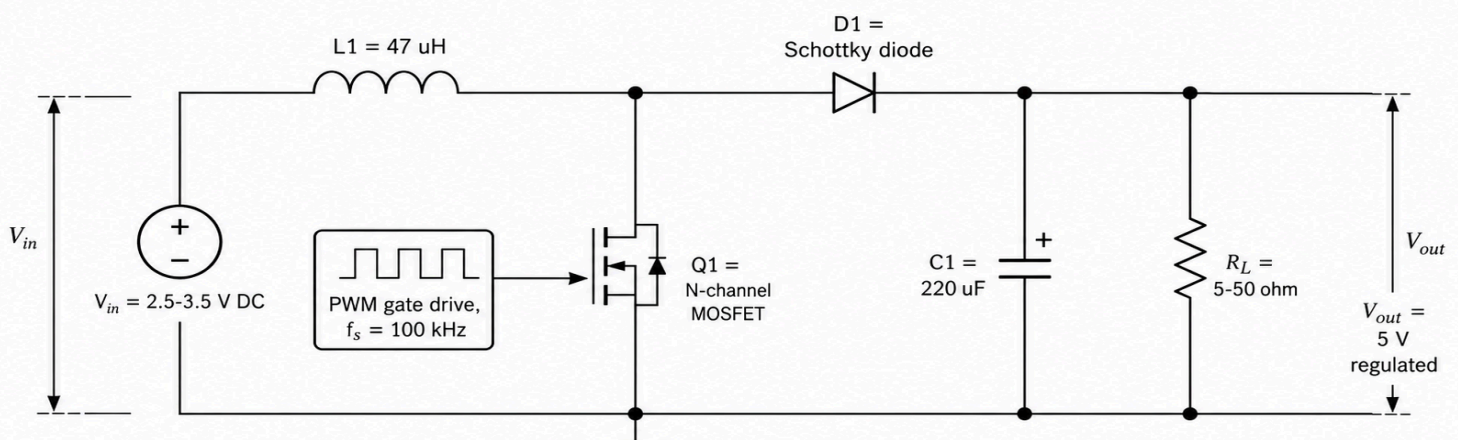
Charger Schematic



Parameter	Value
Input Voltage	5 Vdc
Output Voltage Range	0 Vdc to 3.5Vdc
Voltage Regulation	5%
Output Current Maximum	0.5 A
Maximum Output Power	1.75 W (at 3.5V, 0.5A)
Test Load Resistances	50 ohm, 100 ohm

Current limiting: if input current $> 0.5A$ Bjt on when $V_E = V_B + 0.7V(V_{R2})$ $V_{R2} = \text{Input current} \times \text{Resistance}$ $0.7 = I_{in} \times (2.2\Omega 3.9\Omega)$ $I_{in} = 0.7 / 1.4 = 0.5A$ $R_3 = \text{Pull up resistor}$	Opamp Loop: LM358 max output = 3.5V according to the datasheet
Voltage feedback divider: $V_{fb} = 1.75V$ $V_{fb} = V_{out} \frac{R_5}{R_4 + R_5}$ $V_{fb} = V_{out} \frac{1k}{2k}$ $V_{fb} = \frac{V_{out}}{2}$ $V_{out} = 3.5V$	Compensation Components: $R_9 = 47k, C_2 = 3.3 \times 10^{-9}$ $f = \frac{1}{2\pi R_9 C_2} \rightarrow \frac{1}{2\pi \times 47k \times 3.3 \times 10^{-9}}$ $f = 1.03kHz$
	Mosfet: Mosfet is on when Source $>$ Gate Source = 5V (4.3 if current limiting) Gate = 3.5V from opamp max output Drain = 3.5V from Vref voltage divider Acts as a variable resistor with 1.5V drop

Preliminary Boost Converter Schematic



Parameter	Value
Output Power	0.5 - 5 W
Output Current	0.1 - 1.0 A
Duty Cycle Range	$D = 0.30 - 0.50$
Minimum Load	5 Ω
Maximum Load	50 Ω
Target Ripple	< 50 mV

→ Boost converter does not create power

ideal converter $P_{in} = P_{out}$

$$\therefore V_{in} I_{in} = V_{out} I_{out}$$

$P_{out} = 5W \rightarrow$ At full output power

$$V_{out} = 5V$$

$$\therefore I_{out} = 1A$$

→ At $V_{in} = 2.5V$ the ideal current is

$$I_{in} = \frac{P_{in}}{V_{in}} = \frac{5}{2.5} ; I_{in} = 2A$$

→ Load resistance

$$P = \frac{V^2}{R} ; R = \frac{V^2}{P}$$

$$R_{load} = \frac{5^2}{1} = 5\Omega$$

for minimum power $R_{load} = \frac{5^2}{0.5} = 50\Omega$

$$5\Omega \leq R_{load} \leq 50\Omega$$

Inductor - current ripple

→ The inductor current does not remain perfectly constant. It rises while the MOSFET is ON and falls while it is OFF.

During the ON interval

$$\text{From } V_L = L \frac{\Delta I_L}{\Delta t} \rightarrow V_{in} = L \frac{\Delta I_L}{DT_s} \therefore \Delta I_L = \frac{V_{in} DT_s}{L}$$

$$\text{since } T_s = \frac{1}{f_s} \rightarrow \Delta I_L = \frac{V_{in} D}{L f_s}$$

↳ this is peak to peak current inductor ripple

And we know $I_{L, \text{average, critical}} = \frac{\Delta I_L}{2}$

→ Relating inductor current to output current

the diode delivers inductor current to the output while MOSFET is off

and we know MOSFET is off for $1-D$

$\therefore$ assuming small ripple $I_{out} = (1-D) I_{L, \text{average}}$

→ At the CCM and DCM boundary $I_{out, \text{critical}} = (1-D) \frac{\Delta I_L}{2}$

$$\text{substitute } \Delta I_L = \frac{V_{in} D}{L f_s} \therefore I_{out, \text{critical}} = (1-D) \frac{V_{in} D}{2 L f_s}$$

calculate inductance:

our output specifications $V_{out} = 5V$, $P_{out} = 0.5-5W$, $f_s = 100kHz$

thus $R_{load} = 5\Omega \rightarrow 50\Omega$

→ now to calc inductance

$$\text{use } L_{min} = \frac{R_{load} D (1-D)^2}{2 f_s}$$

substitute max $R_{load} (50\Omega)$

and $D = \frac{1}{3}$ and $f_s = 100kHz$

$$50 \left(\frac{1}{3}\right) \left(1 - \frac{1}{3}\right)^2 \approx 37.0 \times 10^{-6} H$$

$$\therefore L_{min} = 2 (100kHz)$$

$$\therefore L_{min} \approx 37 \mu H$$

→ we should select $47 \mu H \rightarrow$ because components have tolerances, real losses change etc, operation vary and calculation is ideal

The **IRF9224N** P-channel MOSFET was selected as the switching device because it provides low conduction losses and simple gate driven requirements in a positive supply system. Compared with a BJT, the MOSFET requires negligible steady-state gate current and is low-resistance minimising voltage-drop, improving efficiency when charging and discharging.

30F Super Capacitor	150F Super Capacitor
Smaller (Portable)	Holds more energy storage (Practical)
Lower Cost (\$2.10 v \$4.94 when purchased in bulk(2640 units))	More representative of a high-capacity power bank
Quicker charging allows for more testing / prototyping	When the load draws bursts of current, the larger capacitor experiences less instantaneous voltage ripple $\Delta V = I \Delta t / C$
Safer - should any shorts or faults occur, the energy is less than that of 150F reducing the severity of potential damage.	For the same load current, the capacitor voltage drops more slowly because. A larger capacitance means a smaller voltage drop per second. $Q = CV = I \Delta t$
	ESR = 100m ohm @1kHz (AC conditions) vs 200m ohm. 150F more efficient as internal heat losses are half that of 30F capacitor. $P = I^2 \times ESR$

The **150F supercapacitor** was selected to maximise stored energy and extend the operating time of the power bank. Although it requires a longer charging period, it better demonstrates the capability of the charging circuit and provides a more practical energy reserve.

P-type equivalent of BD179 BJT was the BD180, however it is discontinued, so we have used a **BD140** in the current limiting circuitry as its specs are acceptable ($1.5A I_c (\text{Max}) \gg 0.5A$).

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Preliminary Boost Converter Image Generated by <https://chatgpt.com/>

Task Division:

High level system schematic - Bruce/Kyle

Part search & Feasibility study - Alex/Bruce

Calculations/diagrams - Saketh/Kyle